

Assessment Schedule – PRACTICE ASSESSMENT C

Calculus: Apply differentiation methods in solving problems (91578)

Evidence Statement

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
ONE (a)	$\frac{dy}{dx} = \frac{1}{2} \times (-12x) \times (6 - 6x)^{-\frac{1}{2}}.$	• Correct derivative.		
(b)	$\frac{dy}{dx} = 2 \left(2 + \frac{1}{2}(2x)(x^2 + 1)^{-\frac{1}{2}} \right) \left(2x + (x^2 + 1)^{\frac{1}{2}} \right).$ When $x = 2$, $\frac{dy}{dx} = 2 \left(2 + \frac{2}{\sqrt{5}} \right) (4 + \sqrt{5}).$	• Correct gradient found, with evidence of derivative.		
(c)	Velocity $v(t) = 22 - 2 \ln(t + 1) \tan(0.05t)$, Deceleration $a(t) = v'(t) = 0 - \frac{2}{t+1} \tan(0.05t) - 2 \ln(t + 1) (0.05 \sec^2(0.05t)).$ $a(15) = -0.634 \text{ m s}^{-2}$, $0.634 < 0.8$, $\therefore$ the train passes safely. $v(15) = 16.834 \text{ m s}^{-1}.$	• Correct expression for acceleration / deceleration, with evidence of derivative.	• Correct conclusion AND correct final speed found, with evidence of derivative.	
(d)	$f'(x) = (3kx^2 + 4x)e^{2x} + 2(kx^3 + 2x^2 + 3)e^{2x},$ $f'(-1) = (3k - 4)e^{-2} + 2(-k + 2 + 3)e^{-2} = (k + 6)e^{-2}.$ $y - y_1 = m(x - x_1),$ $y - 0 = m(x + 1),$ $y = (k + 6)e^{-2} \times (x + 1),$ $11e^{-2} = (k + 6)e^{-2} \times (0 + 1),$ $k = 5.$ When $k = 5, f'(-1) = 11e^{-2} (\frac{11}{e^2}).$	• Correct expression for $f'(x).$	• Correct value of k and gradient found, with evidence of derivative.	
(e)	$\frac{dx}{dt} = -81 \sin t \cos^2 t,$ $\frac{dy}{dt} = 81 \cos t \sin^2 t,$ $\frac{dy}{dx} = -\frac{81 \cos t \sin^2 t}{81 \sin t \cos^2 t} = -\frac{\sin t}{\cos t}.$ The equation of the tangent: $y - 27 \sin^3 t = -\frac{\sin t}{\cos t} (x - 27 \cos^3 t),$ $y \cos t + x \sin t = 27 \sin t \cos t (\cos^2 t + \sin^2 t),$ Since $\cos^2 t + \sin^2 t = 1,$ $y \cos t + x \sin t = 27 \sin t \cos t.$ Point A $(27 \cos t, 0),$ Point B $(0, 27 \sin t).$ $\sqrt{AB} =$ $\sqrt{(27 \cos t - 0)^2 + (0 - 27 \sin t)^2},$ $= \sqrt{27^2(\cos^2 t + \sin^2 t)} = \sqrt{27^2},$ $\sqrt{AB} = 27.$	• Correct expression for $\frac{dy}{dx}.$	• Equation of the tangent found, with evidence of derivative.	E7 (T1) Correct expression for $\sqrt{AB}$ found, but $\cos^2 t + \sin^2 t = 1$ is not applied OR Correct solution, but with one minor error. E8 (T2) Complete proof, with evidence of a calculus method.

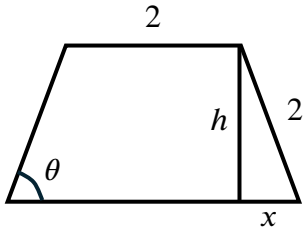
NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
TWO (a)	$\frac{dy}{dx} = \frac{(x^2+1)\left(\frac{-2x}{1-x^2}\right) - (\ln(1-x^2)(2x))}{(x^2+1)^2}.$	• Correct derivative.		
(b)	$F'(t) = 20(2t + 5)^{\frac{1}{2}} + 20t(2t + 5)^{-\frac{1}{2}}.$ $F'(10) = 140$ thousand dollars per month.	• Correct rate of increase, with evidence of derivative.		
(c)	$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt},$ $A = 4\pi r^2, \frac{dA}{dr} = 8\pi r,$ $\therefore -12\pi = 8\pi r \times \frac{dr}{dt},$ $\frac{dr}{dt} = -\frac{3}{2r}.$ $V = \frac{4}{3}\pi r^3, \frac{dV}{dr} = 4\pi r^2.$ $\frac{dV}{dt} = \frac{dV}{dr} \times \frac{dr}{dt} = 4\pi r^2 \times \left(-\frac{3}{2r}\right),$ $\frac{dV}{dt} = -6\pi r.$ When $r = 6, \frac{dV}{dt} = -36\pi$ $\approx -113.1 \text{ cm}^3 \text{ min}^{-1}.$	• Correct expression for $\frac{dV}{dr}$ and $\frac{dr}{dt}.$	• Correct rate of change found, with evidence of derivative.	
(d)	When $x = 2, y_1 = 7,$ Point $P(2,7).$ $\frac{dy_1}{dx} = \frac{2(x-1) - (2x+3)}{(x-1)^2},$ $\frac{dy_1}{dx} = -\frac{5}{(x-1)^2}.$ When $x = 2, \frac{dy_1}{dx} = -5,$ Equation of $L: y = -5x + 17.$ Hence $-5x + 17 = x^2 - x - 4,$ $(x+7)(x-3) = 0,$ $x = -7$ or $x = 3,$ Since we're looking for a positive x -coordinate, $x = 3.$ When $x = 3, y = 2,$ Coordinates of $Q: (3, 2).$	• Correct expression for $\frac{dy_1}{dx}.$	• Correct coordinates found, with evidence of derivative.	

(e)	Let h be the depth of the juice, $V_{\text{juice}} = V_{\text{cylinder to depth } h} - V_{\text{submerged cone}}.$ $V_{\text{cylinder}} = \pi(3r)^2 h = 9\pi r^2 h.$ $V_{\text{total cone}} = \frac{1}{3}\pi r^2(4r) = \frac{4}{3}\pi r^3,$ Let the radius of the cone at the liquid surface be x, By similar triangles, $\frac{x}{r} = \frac{4r-h}{4r}$, $x = r - \frac{h}{4}$, $V_{\text{top cone}} = \frac{1}{3}\pi\left(r - \frac{h}{4}\right)^2(4r - h),$ $= \frac{4}{3}\pi\left(r - \frac{h}{4}\right)^3,$ $V_{\text{submerged cone}} = \frac{4}{3}\pi r^3 - \frac{4}{3}\pi\left(r - \frac{h}{4}\right)^3.$ $V_{\text{juice}} = 9\pi r^2 h - \frac{4}{3}\pi r^3 + \frac{4}{3}\pi\left(r - \frac{h}{4}\right)^3,$ $\frac{dV}{dh} = 9\pi r^2 - \pi\left(r - \frac{h}{4}\right)^2. \quad (1)$ When the depth is halfway up (i.e. $h = 2r$), $\frac{dV}{dh} = 9\pi r^2 - \pi\left(r - \frac{2r}{4}\right)^2,$ $\frac{dV}{dh} = \frac{35\pi r^2}{4}.$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt},$ Given $\frac{dV}{dt} = 10$, $\frac{dh}{dt} = \frac{4}{35\pi r^2} \times 10 = \frac{8}{7\pi r^2} \text{ cm s}^{-1}. \quad (2)$ Alternative method: $dV = A \times dh,$ $\frac{dV}{dh} = \text{Area of cylinder} - \text{Area of cone slice},$ $\frac{dV}{dh} = \pi(3r)^2 - \pi\left(r - \frac{h}{4}\right)^2,$ Substitute $h = 2r$, $\frac{dV}{dh} = 9\pi r^2 - \pi\left(r - \frac{2r}{4}\right)^2 = \frac{35\pi r^2}{4},$ $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt},$ Given $\frac{dV}{dt} = 10$, $\frac{dh}{dt} = \frac{4}{35\pi r^2} \times 10 = \frac{8}{7\pi r^2} \text{ cm s}^{-1}.$	• Finds $\frac{dV}{dh}$ - evidence of reaching stage (1), but with one minor error.	• Correct expression for $\frac{dV}{dh}$ - evidence of reaching stage (1).	E7 (T1) Correct solution, but with one minor error. E8 (T2) Correct rate of increase found / reaching stage (2), with evidence of a calculus method.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Q	Expected Coverage	Achievement (u)	Achievement with Merit (r)	Achievement with Excellence (t)
THREE (a)	(i) $x = -2, x = 2, x = 8$ (ii) $x = -0.5, 4.5 < x < 7, 7 < x < 8, x = 9$ (iii) Does not exist	• TWO out of three responses correct.		
(b)	$\frac{dx}{dt} = 2 \sec(2t) \tan(2t),$ $\frac{dy}{dt} = -2t \sin(t^2),$ $\frac{dy}{dx} = \frac{-2t \sin(t^2)}{2 \sec(2t) \tan(2t)}.$	• Correct expression for $\frac{dy}{dx}.$		
(c)	$f'(x) = \frac{\ln(x)-x-\frac{1}{x}}{\ln^2(x)} = \frac{\ln(x)-1}{\ln^2(x)}.$ $f''(x) = \frac{\frac{\ln^2(x)}{x} - \frac{2\ln(x)}{x}(\ln(x)-1)}{\ln^4(x)},$ $= \frac{\ln^2(x)-2\ln^2(x)+2\ln(x)}{x\ln^4(x)},$ $= \frac{-\ln(x)+2}{x\ln^3(x)}.$ To find points of inflection, $f''(x) = 0.$ Hence $-\ln(x) + 2 = 0,$ $\ln(x) = 2,$ $x = e^2.$	• Correct expression for $f'(x).$	• Correct x -coordinate found, with evidence of derivative.	
(d)	$y = \frac{(e^x+e^{-x})(e^x-e^{-x})}{4}.$ $\frac{dy}{dx} = \frac{(e^x-e^{-x})^2+(e^x+e^{-x})^2}{4},$ $\frac{dy}{dx} = \frac{e^{2x}-1+e^{-2x}+e^{2x}+1+e^{-2x}}{4},$ $\frac{dy}{dx} = \frac{2e^{2x}+2e^{-2x}}{4} = \frac{e^{2x}+e^{-2x}}{2},$ $f(2x) = \frac{e^{2x}+e^{-2x}}{2},$ Hence $\frac{dy}{dx} = f(2x),$ as required.	• Correct expression for $\frac{dy}{dx}.$ Does not need to be simplified.	• Complete proof, with evidence of derivative.	

(e)	 $\sin(\theta) = \frac{h}{2}, h = 2 \sin(\theta).$ $\cos(\theta) = \frac{x}{2}, x = 2 \cos(\theta).$ Area $A = \frac{1}{2}(a + b)h,$ $= \frac{1}{2}(2 + 2 + 4 \cos(\theta))(2 \sin(\theta)),$ $= 4 \sin(\theta) + 4 \cos(\theta) \sin(\theta).$ Applying double angle rule, $A = 4 \sin(\theta) + 2 \sin(2\theta).$ $\frac{dA}{d\theta} = 4 \cos(\theta) + 4 \cos(2\theta).$ To find maximum area, $\frac{dA}{d\theta} = 0,$ so $4 \cos(\theta) + 4 \cos(2\theta) = 0,$ $\cos(\theta) + \cos(2\theta) = 0.$ Applying double angle rule, $\cos(\theta) + 2 \cos^2(\theta) - 1 = 0,$ $(2 \cos(\theta) - 1)(\cos(\theta) + 1) = 0,$ $\cos(\theta) = \frac{1}{2}$ or $\cos(\theta) = -1.$ If $\cos(\theta) = \frac{1}{2}, \theta = \frac{\pi}{3}.$ $\cos(\theta) = -1$ does not give maximum area. $\frac{d^2A}{d\theta^2} = -4 \sin(\theta) - 8 \sin(2\theta).$ When $\theta = \frac{\pi}{3},$ $\frac{d^2A}{d\theta^2} = -4 \sin\left(\frac{\pi}{3}\right) - 8 \sin\left(\frac{2\pi}{3}\right) < 0.$ Therefore $\theta = \frac{\pi}{3}$ maximises the area and the area is indeed a maximum.	• Correct expression for $\frac{dA}{d\theta}.$	• Correct expression for $\frac{dA}{d\theta}$ AND Sets $\frac{dA}{d\theta} = 0$ AND Evidence of reaching $\cos(\theta) + \cos(2\theta) = 0.$	E7 (T1) Correct value of θ found, with evidence of a calculus method, but the maximum area is NOT justified OR Correct solution, but with one minor error. E8 (T2) Correct value of θ found, with full justification and evidence of a calculus method.
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NØ	N1	N2	A3	A4	M5	M6	E7	E8
No response; no relevant evidence.	ONE partial solution.	1u	2u	3u	1r	2r	T1	T2

Cut Scores (2025)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 13	14 – 19	20 – 24

Cut Scores (2024)

Not Achieved	Achievement	Achievement with Merit	Achievement with Excellence
0 – 7	8 – 12	13 – 18	19 – 24